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Abstract

Managing pocket money effectively is a major challenge for many students, often leading to overspending, poor budgeting habits, and lack of financial awareness. This project, Pocket Money Analyzer, aims to provide an intelligent and user-friendly system that helps students track their daily expenses, analyze spending patterns, and improve their financial decision-making.

The system collects expense data entered by users, categorizes it into predefined spending groups, and generates visual summaries such as pie charts, bar graphs, and monthly trend lines. Based on spending behavior, the analyzer provides personalized insights and savings recommendations to encourage better money management.

The project focuses on simplicity, accessibility, and practical usefulness by implementing a lightweight web/mobile-based interface. The results demonstrate that visual feedback and structured financial tracking can significantly improve budgeting habits among students.

Keywords: Pocket money, expense tracking, budgeting, financial awareness, student finance, spending analysis.

CHAPTER one

Introduction

Managing personal finances has become an essential skill in modern student life. Many students receive a fixed amount of pocket money every month, but a large number struggle to control how they spend it. Without proper planning or tracking, they often face situations where money runs out before the month ends, leading to borrowing, unnecessary stress, and poor financial habits. These problems gradually develop into long-term issues, such as careless spending behavior and lack of savings discipline, which can affect financial stability later in life.

Traditionally, students try to remember their expenses mentally or write them down in notebooks. However, these methods are often inaccurate, inconsistent, and difficult to maintain. Students may forget to record small daily expenses such as snacks, transportation, or mobile recharges, which gradually add up to a significant amount. As a result, they fail to understand where most of their money is going and are unable to identify unnecessary spending. Mobile calculators or note-taking apps provide some help, but they do not offer meaningful analysis or guidance for better money management.

With the growing use of smartphones and web-based tools, technology provides an effective way to solve this problem. Digital systems can automatically organize data, classify expenses, and provide visual insights that are easy to understand. By analyzing spending patterns, students can learn to make informed decisions, plan their monthly budgets, and develop better financial habits. Expense tracking applications exist, but many of them are complex, filled with advertisements, require paid subscriptions, or demand bank account access, which makes students hesitant to use them.

To address these concerns, this project introduces “Pocket Money Analyzer,” a simple, user-friendly system designed specifically for students. The main purpose of this project is to help students record their daily expenses and analyze their spending behavior through clear visual representations such as pie charts, bar graphs, and monthly trend lines. Unlike general financial apps, this system focuses on pocket money usage rather than full banking transactions, making it safer, easier, and more relevant for student needs.

The Pocket Money Analyzer allows users to enter their expenses manually and categorizes them into common spending areas, such as food, travel, entertainment, and stationery. It then processes this information and generates useful insights, including total monthly spending, highest spending category, and potential savings opportunities. Through this approach, the system aims to encourage responsible financial behavior and improve awareness of money management among students.

In addition to analytical features, the project emphasizes accessibility and practicality. The system is designed as a lightweight web application with potential support for mobile devices, ensuring that students can access it anytime and anywhere. By providing a clean interface, fast performance, and simple interaction, the Pocket Money Analyzer becomes an effective tool for everyday use.

Overall, this project combines basic data processing, visualization techniques, and user-centered design to create a practical solution for student financial management. By developing the habit of tracking expenses and understanding spending patterns, students can significantly improve their budgeting skills and financial discipline. The Pocket Money Analyzer offers a meaningful step toward

promoting financial awareness and helping students develop responsible money-handling habits that can benefit them throughout their lives.

1.1 Motivation

Pocket money plays an important role in the daily life of students. It is often the first financial resource that young individuals manage on their own, without direct supervision from parents or guardians. This period forms the foundation of their financial habits, which can influence their behavior in adulthood. However, many students lack awareness about how to properly manage their pocket money, leading to careless spending, impulsive purchases, and an inability to save. As a result, they frequently experience situations where their money finishes earlier than expected, causing frustration and dependency on others.

Studies and everyday observations show that most students do not maintain any record of their spending. They may remember major expenses, but small daily purchases such as snacks, bus fare, online subscriptions, or mobile recharges are often ignored. These small transactions accumulate over time and form a major part of monthly expenses. Without clear tracking, students are unable to identify where their money is being spent and which areas need improvement. This lack of financial visibility prevents them from developing responsible money-handling habits.

Traditional methods such as writing expenses in notebooks or mentally remembering them are unreliable and difficult to maintain. Students usually lose interest or forget to update these records regularly. Moreover, manual calculations for monthly totals, category-wise spending, or savings are time-consuming and prone to errors. As a result, students remain unaware of their spending patterns and continue making the same financial mistakes every month.

With the increasing availability of smartphones and internet access, technology offers a powerful opportunity to support students in managing their money more effectively. Digital tools can automatically organize expenses, generate summaries, and provide visual insights that are easy to understand. However, most existing financial apps are designed for adults and require bank account access, credit card linking, or subscription payments. Many are filled with advertisements or complex features that make them difficult for students to use. Due to privacy concerns and lack of relevance to student needs, these applications are not widely adopted in the student community.

This situation highlights the need for a simple, secure, and student-focused solution that helps young users track their pocket money without complexity. A system that allows manual input, performs automatic analysis, and presents results in an easy graphical format can encourage students to become more financially aware. By understanding their spending habits, students can plan better, avoid unnecessary purchases, and develop savings discipline.

The motivation behind the Pocket Money Analyzer project is to bridge this gap by providing a lightweight, user-friendly platform dedicated to students. The goal is to promote financial awareness, help users manage their pocket money more responsibly, and build stronger financial habits that will benefit them in the future. By combining technology with practical daily needs, this project aims to support students in becoming more independent, organized, and financially confident.

1.2 Problem Definition

Students often struggle to manage their pocket money effectively due to the absence of a proper system for recording and analyzing their daily expenses. The key problems include:

• **Lack of structured tracking**
– Students do not maintain a regular record of their spending
– Small daily expenses are ignored or forgotten
– Leads to unclear financial understanding

• **Uncontrolled spending habits**
– Impulsive or unnecessary purchases go unnoticed
– Money finishes earlier than expected
– Students become dependent on others for additional funds

• **Ineffective traditional methods**
– Mental tracking and handwritten notes are inconsistent
– Manual calculation of totals and categories is time-consuming
– Information becomes incomplete and inaccurate

• **No spending analysis**
– Students cannot identify where most of their money is going
– No category-wise breakdown (food, travel, entertainment, etc.)
– No awareness of wasteful spending areas

• **Lack of financial planning**
– Students do not set budgets or savings goals
– No understanding of how to allocate money properly

• **Unsuitable existing digital tools**
– Most expense apps are designed for adults
– Require bank linking, credit card access, or subscriptions
– Interfaces are complex and filled with advertisements

• **Privacy and security concerns**
– Students hesitate to share financial details with applications
– Fear of data misuse or unauthorized access

Because of these issues, students remain unaware of their spending patterns and are unable to develop responsible financial habits. There is a clear need for a simple, secure, and student-friendly system that allows easy expense recording, automatic categorization, and meaningful analysis.

1.2.1 Challenges with Existing Money Management Methods

Students face several difficulties when managing their pocket money using traditional or commonly available methods. The major challenges include:

• **Memory-based tracking**
– Students try to remember expenses mentally

– Small daily purchases are easily forgotten
– Leads to inaccurate understanding of total spending

• **Irregular handwritten records**
– Maintaining notebooks requires discipline
– Students often forget to update entries
– Manual calculations become tiring and error-prone

• **Lack of categorization**
– Traditional methods do not separate expenses into categories such as food, travel, or entertainment
– Students cannot identify which area consumes most of their money

• **No visual analysis**
– Existing methods do not provide charts or graphs
– Students fail to recognize spending patterns or trends

• **Complex financial applications**
– Many available apps are designed for adults
– Include features like bank linking, credit cards, and investment tracking
– Difficult for students to use and understand

• **Privacy concerns**
– Students hesitate to share personal financial information
– Bank integration increases security risks

• **Advertisements and paid features**
– Free apps contain distracting ads and limited functionality
– Premium versions may not be affordable for students

• **Lack of motivation to continue**
– Traditional methods feel boring or time-consuming
– Students quickly stop maintaining records

These challenges clearly show that existing money management methods are not suitable for student needs and do not help them build effective financial habits.

1.2.2 Need for Financial Awareness Among Students

Financial awareness is an essential skill that students must develop during early stages of life, as it forms the foundation of their future money-handling behavior. The need for financial awareness among students arises due to the following reasons:

• **Limited financial resources**
– Students usually receive a fixed amount of pocket money
– Poor planning causes money to run out quickly
– Leads to financial stress and dependency

• **Lack of spending visibility**
– Students do not know where their money is going
– Small daily expenses accumulate unnoticed
– No clear understanding of spending patterns

• **Absence of budgeting habits**
– Students rarely create spending plans
– No allocation for essential and non-essential expenses
– No habit of setting financial limits

• **Uncontrolled and impulsive spending**
– Purchases influenced by friends or trends
– Frequent spending on food, entertainment, and online services
– No evaluation of financial priorities

• **Lack of savings discipline**
– Students do not consider saving a part of their pocket money
– No future financial goals or emergency funds
– Missed opportunity to build responsible habits early

• **Limited financial education**
– Schools and colleges rarely teach personal money management
– Students learn through trial and error
– Leads to long-term financial problems in adulthood

• **Influence of digital spending**
– Online payments, UPI, and subscriptions make spending easier
– Reduced sense of money value
– Increased chances of overspending

• **Long-term impact on life**
– Poor financial habits in youth continue into adulthood
– Difficulty managing salaries, bills, and savings later
– Higher risk of debt and financial instability

These points clearly show that financial awareness is not just helpful but necessary for students. Developing financial management skills at an early age can improve their decision-making, promote responsible spending, and build strong financial habits for the future.

1.2.3 Technical Direction of the Proposed Solution

To overcome the problems identified in existing money management methods, this project follows a clear technical direction focused on simplicity, accessibility, and effective data analysis. The proposed solution aims to provide a structured system that helps students track and understand their spending through the following technical approach:

• **Manual expense input system**
– Users enter their daily expenses manually
– Eliminates the need for bank linking or automated transactions
– Ensures privacy and data control

• **Automatic categorization of expenses**
– System assigns expenses to categories such as food, travel, entertainment, and stationery
– Helps users understand major spending areas
– Reduces effort required from students

• **Data storage and organization**
– Expenses are stored in a structured format
– Enables easy retrieval and analysis
– Supports month-wise tracking

• **Visual data representation**
– Pie charts, bar graphs, and trend lines display spending patterns
– Makes financial information easy to understand
– Encourages user engagement

• **Spending analysis module**
– Calculates totals and category-wise spending
– Identifies highest spending areas
– Highlights unnecessary expenses

• **Savings recommendation system**
– Suggests possible savings based on spending patterns
– Encourages better financial habits
– Provides personalized insights

• **Lightweight web-based interface**
– Can be accessed easily through browsers
– Fast and simple to use
– Suitable for students with basic devices

• **Future mobile support**
– Design allows easy extension to mobile applications
– Responsive interface for smartphones
– Increases accessibility and convenience

This technical direction ensures that the Pocket Money Analyzer remains simple, secure, and effective while providing meaningful financial insights tailored to student needs.

1.2.4 Deployment Considerations (Web/App)

Deploying the Pocket Money Analyzer requires careful planning to ensure that the system remains lightweight, accessible, and user-friendly for students. Since the target users primarily rely on smartphones and basic laptops, the deployment environment must support smooth performance without demanding high system resources. The project focuses on a web-based deployment first, with the flexibility to extend into mobile platforms in the future.

To achieve this, the system is designed to run efficiently on common browsers without requiring installation or complex setup. A web application allows students to access the tool from anywhere, making it more convenient for regular usage. However, simply hosting a web interface is not enough; the system must also consider device compatibility, network availability, and user experience.

Key deployment considerations include:

• **Lightweight performance**
– The system must load quickly on low-end devices

– Minimal processing should occur on the client side
– Avoid heavy libraries that increase loading time

• **Responsive interface design**
– Layout adapts to different screen sizes
– Suitable for smartphones, tablets, and laptops
– Ensures comfortable usage without zooming or scrolling issues

• **Offline usability (future enhancement)**
– Students may not always have stable internet access
– Local storage or caching can allow temporary offline usage
– Improves reliability in real-life student environments

In addition to web deployment, mobile support is an important consideration. Many students prefer mobile applications for daily use because they are more accessible and provide faster interaction. Although the current version focuses on the web, the system design allows easy conversion into an Android application using the same backend logic.

• **Potential Android integration**
– Use of frameworks such as Flutter or React Native
– Integration of the existing web logic with mobile UI
– Better accessibility and long-term usability

Security and privacy are also critical factors. Since students manually enter their expenses, the system must ensure that the data is stored securely and remains private. Unlike banking applications, this system does not require sensitive financial information, which reduces risk; however, basic protection measures remain necessary.

• **Data privacy considerations**
– Protect user-entered information
– Avoid unnecessary data collection
– Provide safe storage mechanisms

Overall, deployment decisions for the Pocket Money Analyzer aim to balance accessibility, performance, and usability. By prioritizing a lightweight web application with potential mobile expansion, the project ensures that students can easily adopt and use the system in their daily lives.

1.2.5 Formal Problem Statement

The primary problem addressed in this project is the lack of a simple, accessible, and student-focused system for managing pocket money effectively. Students often fail to track their daily expenses, leading to unclear spending patterns, poor budgeting habits, and financial instability. Existing methods such as mental tracking, handwritten records, or complex financial applications do not provide structured analysis, visual insights, or a user-friendly experience suitable for student needs. Therefore, there is a need for a lightweight and secure solution that allows students to manually record their expenses, automatically categorize them, and generate meaningful visual summaries to support better financial decision-making.

Formally, the problem can be stated as follows:

"To design and develop a lightweight and user-friendly expense analysis system that enables students to record, categorize, and visualize their pocket money usage, providing meaningful insights and promoting responsible financial habits through a web-based platform with future mobile support."

1.2.6 Expected Outcomes of Problem Resolution

By addressing the problems related to poor pocket money management among students, this project aims to create a meaningful and practical impact. The Pocket Money Analyzer is expected to not only help students track their expenses but also improve their financial behavior through clear insights and simplified budgeting practices. With a structured system for recording and analyzing spending, students will gain better control over their finances and develop responsible habits that can benefit them throughout their lives.

The expected outcomes of this project include:

• **Improved spending awareness**
– Students will clearly understand where their money is being spent
– Small daily expenses will become visible and measurable

• **Better budgeting habits**
– Users will be able to plan their monthly expenses
– More control over essential and non-essential spending

• **Identification of unnecessary expenses**
– System highlights categories with excessive spending
– Helps students reduce wasteful purchases

• **Increased savings potential**
– Students can set informal savings goals
– Visual feedback encourages saving behavior

• **User engagement through visuals**
– Charts and graphs make financial data easy to interpret
– Enhances motivation to continue using the system

• **Accessible financial tool**
– Lightweight web application usable on basic devices
– No need for bank linking or sensitive information

• **Foundation for mobile expansion**
– System design supports future Android app development
– Increases accessibility and daily usage

Overall, the successful implementation of the Pocket Money Analyzer is expected to help students gain financial awareness, develop better money management skills, and build long-term responsible financial habits in a simple and engaging manner.

1.3 Objectives

The main objective of this project is to design and develop a simple and effective system that helps students manage their pocket money more responsibly. By providing an organized platform to record expenses and visualize spending patterns, the Pocket Money Analyzer aims to improve financial awareness and encourage better budgeting habits among students. The project focuses on usability, accessibility, and practical usefulness to ensure that students can easily adopt and benefit from the system in their daily lives.

To achieve this goal, the project defines the following specific objectives:

1. **To develop a user-friendly system for recording daily expenses**
   that allows students to manually enter their spending with minimum effort.

2. **To categorize expenses automatically**
   into common spending groups such as food, travel, entertainment, and stationery.

3. **To analyze spending patterns**
   by calculating total expenses and identifying major spending categories.

4. **To provide visual representations of financial data**
   using pie charts, bar graphs, and trend lines for easy understanding.

5. **To offer basic savings recommendations**
   based on spending behavior to promote better financial decisions.

6. **To design a lightweight web-based interface**
   that works smoothly on basic devices and common browsers.

7. **To ensure privacy and data security**
   by avoiding bank integration and protecting user-entered information.

These objectives collectively aim to create a practical and accessible financial tool that supports students in developing responsible money management habits.

1.4 Scope of the Project

The scope of this project focuses on developing a software-based solution that helps students manage their pocket money by tracking and analyzing their daily expenses. The Pocket Money Analyzer is designed as a lightweight web application that allows users to manually enter their spending, categorize expenses, and view meaningful visual summaries. The project emphasizes simplicity, accessibility, and practicality, ensuring that students can use the system easily without requiring advanced financial knowledge or complex setup procedures.

This project primarily covers:

• **Expense recording and storage**
– Manual input of daily expenses
– Structured storage for easy retrieval and analysis

• **Automatic categorization of expenses**
– Assigning spending to predefined categories
– Helping users understand major spending areas

• **Spending pattern analysis**
– Calculation of total and category-wise expenses
– Identification of highest spending categories

• **Data visualization**
– Pie charts, bar graphs, and trend lines
– Clear and engaging presentation of financial information

• **Savings recommendations**
– Basic suggestions to improve financial habits
– Encouragement toward responsible spending

• **Web-based deployment**
– Accessible through common browsers
– Lightweight performance on basic devices

In order to maintain focus and feasibility, the project does not include advanced financial features such as bank account integration, automated transaction imports, investment tracking, or professional budgeting tools. These features require sensitive financial data and complex security measures, which are beyond the current scope and not essential for student-focused pocket money management.

The project also limits its scope to individual usage rather than multi-user systems or cloud-based financial platforms. However, the design allows for future expansion into mobile applications and additional features based on user feedback and requirements.

Overall, the scope of the Pocket Money Analyzer ensures the development of a practical and student-oriented financial tool that provides essential expense tracking and analysis features while remaining simple, secure, and easy to use.

CHAPTER TWO

Dataset / Data Collection

2.1 Source

The dataset used in this project is sourced from **Kaggle**, a widely recognized online platform that provides high-quality datasets for data science and machine learning projects. Kaggle is commonly used by researchers, students, and professionals due to its large collection of publicly available datasets, reliable hosting, and community support. The dataset selected for this project contains expense-related information that reflects common spending categories such as food, transportation, entertainment, and personal items, which are relevant to student pocket money usage.

Kaggle provides structured datasets in formats such as CSV or Excel, making them easy to import and process for analysis. The dataset chosen for this project offers a realistic representation of daily expenses and includes entries with date, amount, and spending category. This structure supports the objective of analyzing spending patterns and categorizing expenses effectively.

The dataset can be accessed through the following platform:

• Kaggle: https://www.kaggle.com

Using a Kaggle dataset ensures that the project is based on reliable and organized data, which enhances the accuracy and credibility of the results.

2.2 Dataset Description

The dataset used in this project contains detailed information about student spending patterns and financial situations. It includes both demographic attributes and multiple categories of expenses, making it highly suitable for analyzing how students manage their pocket money. The dataset consists of several columns that represent student characteristics such as age, gender, major, and year in school, along with financial details including monthly income and financial aid. This structure helps in understanding not only how much students spend but also the factors that may influence their spending behavior.

Each record in the dataset represents an individual student and includes the total amount spent in different expense categories. These categories cover essential and non-essential spending areas, allowing meaningful comparisons and analysis of financial priorities. The dataset provides a realistic representation of student expenses and supports the objectives of this project by enabling accurate calculation of spending patterns and potential savings.

The main features included in the dataset are:

• **Demographic information**
– Age
– Gender

– Major/Field of study
– Year in school (Freshman, Junior, Senior, etc.)

• **Financial background**
– Monthly income
– Financial aid received

• **Expense categories**
– Tuition
– Housing
– Food
– Transportation
– Books and supplies
– Entertainment
– Personal care
– Technology
– Health and wellness
– Miscellaneous spending

• **Payment methods**
– Credit/Debit Card
– Cash
– Mobile payment apps

The dataset is stored in a structured tabular format (CSV), making it easy to process and analyze using data analysis tools. The presence of multiple expense-related columns allows the system to categorize and calculate spending totals accurately, which is essential for generating visual insights and recommendations.

Overall, this dataset provides a rich and comprehensive foundation for understanding student spending habits and supports the development of an effective pocket money analysis system.

2.3 Why This Dataset?

This dataset was selected because it provides a realistic and detailed representation of how students manage their finances, making it highly suitable for the Pocket Money Analyzer project. Unlike generic expense datasets that focus on adult spending or general household budgets, this dataset specifically targets students, which aligns directly with the objective of analyzing pocket money usage among young individuals. It includes both financial inputs, such as monthly income and financial aid, and multiple expense categories that reflect real student lifestyles.

Another major advantage of this dataset is its structured format. The data is organized in a clear tabular layout with separate columns for each spending category, which simplifies the process of categorization and analysis. The presence of demographic information also allows deeper insights into how factors such as age, major, or preferred payment method may influence spending behavior. This makes the dataset not only useful for basic analysis but also valuable for identifying patterns and trends within different student groups.

Key reasons for choosing this dataset include:

• **High relevance to the project**
– Focuses specifically on student spending
– Includes common pocket money expense categories

• **Comprehensive financial coverage**
– Contains income, financial aid, and multiple spending areas
– Supports complete financial behavior analysis

• **Structured and clean format**
– CSV format easy to process and import
– Well-labeled columns suitable for categorization

• **Realistic spending patterns**
– Reflects actual student lifestyle expenses such as food, transportation, entertainment, and personal care
– Provides meaningful data for visualization

• **Supports visualization and insights**
– Enables creation of pie charts, bar graphs, and trend analysis
– Helps identify major spending categories

Because of these characteristics, this dataset provides a strong foundation for developing an effective pocket money analysis system. It supports accurate calculation, meaningful interpretation, and clear visualization of student spending behavior, making it an ideal choice for this project.

2.4 Limitations of the Dataset

Although the dataset provides valuable information about student spending habits, it also has certain limitations that may affect the accuracy and generalization of the analysis. These limitations must be considered while interpreting the results of the Pocket Money Analyzer, as they may influence the conclusions drawn from the data. The dataset mainly reflects structured spending categories but does not fully represent every aspect of student financial behavior.

The major limitations include:

• **Lack of daily transaction records**
– The dataset contains monthly spending totals instead of day-to-day expenses
– Prevents detailed analysis of daily spending patterns or impulse purchases

• **Presence of major expenses**
– Includes large costs such as tuition and housing
– These expenses may overshadow smaller pocket money spending categories
– Makes it difficult to focus only on pocket money usage

• **Synthetic or self-reported data risk**
– Data may not represent actual real-world spending
– Students may underestimate or overestimate certain expenses

• **Limited behavior-related information**
– Does not include motivations behind spending
– No data on emotional or social factors influencing purchases

• **No geographic or cultural context**
– Spending habits may vary by region or country
– Dataset does not specify location, reducing generalization

• **Payment method bias**
– Dataset includes only a few payment categories
– Does not represent all possible transaction methods used by students

In addition, the dataset does not track changes over time for individual students, which limits the ability to observe spending trends or improvements in financial behavior. Because of these limitations, the results of the analysis should be interpreted as general spending patterns rather than precise financial tracking.

Despite these constraints, the dataset remains highly useful for this project, as it provides structured financial information that supports meaningful analysis and visualization of student spending.

2.5 Dataset Preparation and Cleaning

Before using the dataset for analysis, several preparation and cleaning steps were carried out to ensure that the data was suitable for processing and visualization. Proper dataset preparation is essential for generating accurate and meaningful results, as raw data often contains unnecessary columns, inconsistent formats, or missing values. Cleaning the dataset also improves system performance and helps maintain reliability during analysis.

In this project, the dataset was first imported in CSV format using data processing tools. The initial inspection showed that the dataset contained multiple columns related to student demographics, financial background, and categorized expenses. An additional index-like column ("Unnamed: 0") was present, which did not contribute any useful information and therefore needed to be removed.

The dataset preparation and cleaning process involved the following steps:

• **Data loading and inspection**
– Imported the CSV file into the system
– Checked column names and data structure
– Identified unnecessary or redundant fields

• **Removal of irrelevant columns**
– Dropped the "Unnamed: 0" column, which served only as an index
– Ensured that only meaningful attributes remained for analysis

• **Handling missing or inconsistent values**
– Verified whether any fields contained blank or null entries
– Ensured that all financial values were numerical and usable for calculations

• **Standardization of data types**
– Confirmed that expense values were stored in numeric format
– Ensured categorical fields such as gender and payment method were consistent

• **Organization of expense categories**
– Grouped related spending columns for easier analysis
– Prepared the data for category-wise visualization

After cleaning, the dataset became more structured and efficient to work with. The removal of unnecessary columns and validation of data formats helped improve the accuracy of calculations and the clarity of visual outputs. This preparation process ensured that the Pocket Money Analyzer could process the dataset smoothly and generate reliable insights about student spending patterns.

2.6 Dataset Justification

The selected dataset is well-suited for this project because it provides comprehensive and structured information about student spending behavior, which directly supports the objectives of the Pocket Money Analyzer. Since the focus of this project is to analyze how students use their pocket money and identify major spending areas, a dataset that specifically represents student financial activities is essential. The Kaggle dataset used in this project contains detailed expense categories and financial background information, making it highly relevant and effective for generating meaningful insights.

Unlike general financial datasets that include household or corporate expenses, this dataset focuses entirely on students, which ensures that the analysis remains accurate and aligned with the target user group. The presence of multiple expense categories such as food, transportation, entertainment, personal care, and miscellaneous spending allows the system to categorize expenses effectively and produce clear visual summaries. This level of detail supports the goals of the Pocket Money Analyzer in identifying spending patterns and providing useful recommendations.

The dataset is justified for use in this project due to the following reasons:

• **Direct relevance to student financial behavior**
– Contains real or realistic student expense patterns
– Matches the target audience of the system

• **Categorized expense structure**
– Multiple spending categories allow detailed analysis
– Supports category-wise visualization and insights

• **Availability of financial background data**
– Monthly income and financial aid help understand spending capacity
– Enables deeper interpretation of financial decisions

• **Clean and organized format**
– CSV structure allows easy processing
– Well-labeled columns improve data handling

• **Supports project objectives**
– Enables calculation of totals and spending trends
– Helps in generating savings recommendations

In addition, the dataset provides enough variation across students to allow meaningful comparisons and general observations about financial habits. Its structure supports both simple and advanced analysis, making it a strong foundation for the development and evaluation of the Pocket Money Analyzer.

Overall, this dataset is academically appropriate, practically relevant, and technically suitable for achieving the goals of the project.

2.7 Expense Categories

For effective analysis and visualization, the Pocket Money Analyzer groups student expenses into clearly defined categories. These categories represent common areas where students spend their money and form the basis for generating meaningful insights. By organizing expenses into specific groups, the system can calculate category-wise totals, identify major spending areas, and highlight opportunities for savings. This categorization also supports the creation of visual charts such as pie charts, bar graphs, and trend lines, making financial information easier to interpret.

The dataset used in this project includes the following expense categories:

• **Housing**
Includes accommodation-related expenses such as rent, hostel fees, or utilities.

• **Food**
Covers daily meals, snacks, beverages, and groceries, which are major components of student spending.

• **Transportation**
Includes travel expenses such as bus fares, fuel, rideshare services, or public transport usage.

• **Entertainment**
Includes leisure activities such as movies, outings, gaming, and streaming services.

• **Personal Care**
Covers hygiene and self-care expenses such as toiletries, grooming, and related products.

• **Technology**
Includes electronic devices, accessories, software, and maintenance costs.

• **Health and Wellness**
Represents medical expenses, fitness-related spending, or health products.

• **Miscellaneous**
Covers additional expenses that do not fall into the categories above, such as gifts or unexpected purchases.

These categories allow the system to perform detailed financial analysis and generate insights that students can easily understand. By identifying how much money is spent in each area, the Pocket Money Analyzer helps students recognize their spending priorities and make more informed financial decisions.

CHAPTER THREE

Methodology

3.1 System Design and Rationale

The Pocket Money Analyzer follows a structured and systematic design to ensure that the system remains simple, fast, and useful for students. The methodology focuses on collecting expense data, organizing it, analyzing spending patterns, and presenting the results through meaningful visualizations. A step-by-step process is used to transform raw data into useful financial insights that can help students improve their money management habits.

The overall system workflow includes four major stages:

• **Data Collection**
User or dataset provides spending information

• **Data Processing and Categorization**
System organizes and groups expenses

• **Analysis and Calculation**
Determines totals, categories, and spending patterns

• **Visualization and Recommendations**
Displays charts and provides savings guidance

This structure ensures that the system handles data efficiently and presents information in a clear and engaging manner.

The system design was chosen based on the following rationale:

• students need a simple and accessible tool
• visual outputs increase understanding and motivation
• manual data entry ensures privacy and avoids bank integration
• web-based deployment increases accessibility

Unlike complex financial systems that rely on automated transaction imports or banking APIs, the Pocket Money Analyzer uses a lightweight approach. This makes the system easier to deploy and safer for students to use, while still providing useful financial insights.

To illustrate the structure, the system can be represented using a **System Architecture diagram** (Figure 3.1), which includes:

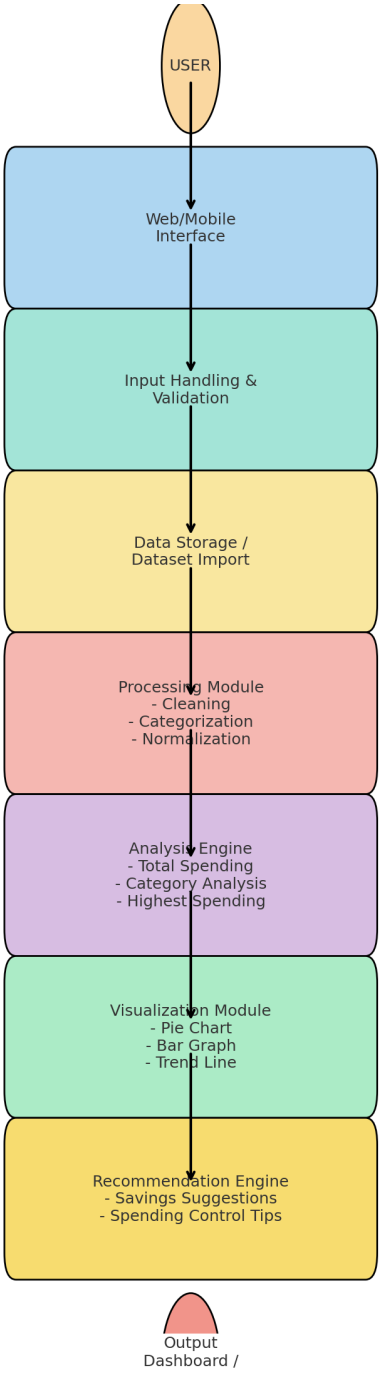
• input module
• processing module
• analysis module
• visualization module
• output interface

Each module works together to convert spending data into meaningful financial information.

This modular design allows future improvements such as:

• mobile app integration
• AI-based recommendations
• automatic data entry features

Overall, the system design ensures that the Pocket Money Analyzer remains practical, efficient, and suitable for everyday student use.



flowchart of the proposed model architecture

3.2 Data Input and Processing Strategy

The Pocket Money Analyzer processes financial information by converting raw expense data into structured and meaningful insights. The system begins by accepting input in the form of student spending records, either manually entered by users or imported from a dataset such as the Kaggle file used in this project. Once the data enters the system, it must be organized and prepared for analysis to ensure accuracy and usefulness.

In the current implementation, data input is handled through a CSV dataset containing multiple financial categories. When deployed as a web application, users will enter their daily expenses manually through an input form. This dual input approach ensures flexibility, allowing the system to operate both with pre-existing datasets and real-time user data.

The processing strategy involves several important steps that transform raw values into categorized, analyzable information:

• **Data validation**
– Checks whether the entered or imported values are valid
– Ensures that numerical fields contain numbers
– Prevents missing or corrupted entries from affecting analysis

• **Expense categorization**
– Assigns each expense to a category such as food, travel, entertainment, and personal care
– Uses column labels in the dataset for automatic grouping
– Supports clearer spending analysis

• **Data normalization**
– Removes unnecessary columns such as the index field
– Ensures consistent formatting across all records

• **Total and category-wise calculations**
– Sums expenses across each category
– Identifies the highest spending area
– Computes total spending for each student

• **Preparation for visualization**
– Data is reorganized into formats suitable for charts and graphs
– Enables creation of pie charts, bar graphs, and trend lines

Unlike traditional financial systems that rely on automated transaction imports, this project intentionally keeps data entry simple and controlled. This approach supports privacy and avoids the need for integrating banking systems, which can introduce security concerns and increase complexity.

By applying this processing strategy, the Pocket Money Analyzer ensures that raw financial data becomes structured, accurate, and ready for meaningful analysis. This step is essential for generating reliable insights and supports the overall goal of helping students understand and improve their spending habits.

3.3 Spending Pattern Analysis

Spending pattern analysis is one of the core functions of the Pocket Money Analyzer, as it helps students clearly understand how their money is being used. After the data has been processed and organized, the system calculates the total spending and identifies how much money is allocated to each expense category. This analysis transforms raw numerical values into meaningful insights that students can interpret easily.

The analysis begins by summing all expense categories for each student, allowing the system to determine overall spending levels. Once the total amount is calculated, the system compares category-wise expenses to identify which areas contribute the most to student spending. This process highlights spending priorities and reveals habits that may not be obvious when looking at individual transactions.

To perform this analysis, the system follows a structured approach:

• **Total spending calculation**
– Adds values across all expense columns
– Determines overall monthly expenditure for each student

• **Category-wise comparison**
– Calculates the percentage of spending in each category
– Identifies the highest and lowest spending areas

• **Trend identification**
– Observes patterns such as consistently high food or entertainment spending
– Helps detect areas where students may overspend

• **Spending behavior interpretation**
– Determines whether spending is balanced or focused on specific categories
– Provides the basis for recommendations

This information is then prepared for visual representation. For example, a **pie chart (Figure 3.4)** can show the proportion of spending in each category, making it easy to identify major expense areas at a glance. A **bar graph (Figure 3.7)** can compare spending between categories, and a **line graph (Figure 3.5)** can illustrate trends over time when multiple monthly records are available.

In the context of the dataset used in this project, analysis reveals that essential expenses such as tuition, housing, and food typically represent the largest portion of student spending. However, categories like entertainment, technology, and personal care also contribute significantly and may indicate areas where students can reduce unnecessary expenditures.

The spending pattern analysis module plays a crucial role in helping students become aware of their financial habits. By visualizing how money is distributed across categories, the system encourages users to reflect on their spending behavior and make more informed decisions. This analytical process supports the main goal of the project by promoting responsible money management.

3.4 Recommendation Logic

The recommendation logic is a crucial component of the Pocket Money Analyzer, as it converts analytical results into practical guidance that students can apply in their daily lives. While understanding spending patterns is important, students benefit more when the system suggests specific actions that can help them improve their financial habits. Therefore, the recommendation module uses category-wise spending analysis to generate simple and meaningful advice tailored to the user's financial behavior.

The recommendation logic follows a rule-based approach that evaluates spending distribution across categories. Instead of using complex machine learning models, the system applies clear and understandable rules that students can easily relate to. This ensures that the recommendations remain transparent, practical, and suitable for a student-focused financial tool.

The system generates recommendations based on the following logic steps:

• **Identification of highest spending category**
– Determines which category consumes the most money
– If the category is non-essential (e.g., entertainment or technology), the system suggests reducing spending

• **Comparison with essential categories**
– Checks whether essential spending such as food and transportation is reasonable
– Highlights imbalance if non-essential spending is higher than essentials

• **Savings potential estimation**
– Calculates how much could be saved by reducing certain expenses
– Suggests a possible savings percentage (e.g., reduce entertainment spending by 10–20%)

• **Spending behavior evaluation**
– Detects patterns such as frequent high spending on food or online services
– Provides targeted advice based on habits

Examples of generated recommendations include:

• “Your entertainment spending is significantly higher than other categories. Reducing it by 15% could increase your savings.”
• “Food expenses represent a large portion of your spending. Consider planning meals to reduce costs.”
• “Technology-related purchases are high this month. Avoid unnecessary gadgets or accessories.”
• “You can save more by limiting miscellaneous spending and tracking smaller purchases.”

In future versions, the recommendation logic can be extended to include:

• machine learning-based prediction models
• personalized budgeting plans
• monthly savings goals
• habit-tracking features

However, the current rule-based system remains effective for providing clear and actionable guidance without increasing complexity.

Overall, the recommendation logic strengthens the purpose of the Pocket Money Analyzer by helping students convert financial insights into positive behavior changes, ultimately promoting responsible money management.

3.5 Visualization Module

The visualization module plays an essential role in the Pocket Money Analyzer, as it converts numerical spending data into clear and easy-to-understand graphical formats. Many students find it difficult to interpret raw financial numbers, but visual representations such as charts and graphs allow them to quickly identify where most of their money is being spent. By presenting information visually, the system increases user engagement and encourages students to reflect on their financial decisions.

The visualization process begins after the system completes category-wise calculations. The processed data is then organized into formats suitable for graphical representation. Visualization not only helps users understand spending distribution but also highlights patterns and trends that may not be obvious from tables or lists.

The module generates the following key visual outputs:

• **Pie Chart (Figure 3.4)**
– Shows the percentage distribution of spending across categories
– Helps users identify major expense areas at a glance
– Useful for understanding overall spending structure

• **Bar Graph (Figure 3.7)**
– Compares spending between categories in terms of absolute values
– Makes it easy to spot categories with unusually high spending
– Effective for identifying non-essential spending areas

• **Line Graph (Figure 3.5)**
– Represents spending trends over time when multiple months are available
– Helps users observe changes in spending behavior
– Useful for long-term financial planning

These visualizations transform the dataset from a simple table of numbers into meaningful financial insights. Students can see instantly whether they are spending too much on food, entertainment, technology, or other categories, which motivates them to adjust their habits.

The visualization module offers additional benefits:

• increases user interest and interaction
• simplifies interpretation for non-technical users
• provides immediate feedback
• supports better decision-making

Unlike traditional text-based reports, graphical outputs make financial information more accessible and easier to remember. This feature is particularly valuable for students who prefer visual learning and quick summaries rather than detailed numerical analysis.

In future versions, the visualization module can be expanded to include:

• interactive dashboards
• monthly comparison charts
• heatmaps of spending frequency
• personalized trend tracking

Overall, the visualization module strengthens the effectiveness of the Pocket Money Analyzer by presenting financial information in a clear, engaging, and user-friendly manner.

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3.7 Limitations of the Methodology

Although the methodology adopted in the Pocket Money Analyzer provides a clear and structured approach for analyzing student spending patterns, it has certain limitations that may affect the depth and accuracy of the results. The system mainly relies on manually entered or pre-structured data, which simplifies processing but restricts the ability to capture real-time financial behavior. Additionally, the rule-based recommendation approach offers practical guidance but lacks personalization at a deeper analytical level.

The major methodological limitations include:

• **Dependence on manual input or static datasets**
– When users enter expenses manually, they may forget to record some transactions
– The system cannot verify the accuracy of entered data
– Real-time spending behavior is not automatically captured

• **Rule-based recommendations**
– Recommendations are generated using simple predefined rules
– The system does not learn from user behavior over time
– Advice may not fully match individual financial priorities

• **Limited trend analysis**
– The methodology works well with monthly totals
– Without continuous or time-series data, long-term behavior cannot be analyzed accurately

• **No behavioral or psychological interpretation**
– The system does not consider emotional or social factors influencing spending
– It cannot detect reasons behind overspending, only numerical outcomes

• **Single-user focus**
– Methodology is designed for individual analysis
– Does not support comparisons between multiple users or groups within the system

In addition, the visualization approach is static, meaning users can view charts but cannot interact with them—for example, filtering categories, adjusting time periods, or exploring detailed spending breakdowns. While this methodology is sufficient for educational and analytical purposes, it limits the system's ability to provide personalized and adaptive financial guidance.

Despite these limitations, the chosen methodology remains appropriate for the current scope of the project. It successfully transforms expense data into clear insights, supports student financial awareness, and aligns with the goal of providing a simple, accessible tool for money management. Future improvements can address these limitations to create a more advanced and interactive system.

CHAPTER FOUR

Implementation

4.1 Environment

The implementation of the Pocket Money Analyzer was carried out using Python in a Jupyter/Google Colab environment. This setup was chosen because it provides an interactive platform for writing, executing, and testing code, making it suitable for data analysis and visualization tasks. Colab also allows the use of Python libraries without requiring complex installation processes, which makes development faster and more accessible.

The implementation environment consists of two main components: hardware support and software tools.

4.1.1 Hardware Environment

Since the project was developed in Google Colab, the hardware resources were provided by cloud computing servers. The system specifications typically included:

• Cloud-based CPU processing
• 12–16 GB RAM (shared environment)
• High-speed storage access
• Internet connectivity for dataset loading and library installation

This environment ensured smooth execution of data processing and visualization tasks without requiring high-end personal hardware.

4.1.2 Software Environment

The software environment included:

• Python programming language
• Google Colab/Jupyter Notebook interface
• CSV dataset imported from Kaggle
• Python libraries for data processing and visualization

The key libraries used in the implementation were:

• Pandas – for dataset loading and manipulation
• NumPy – for numerical operations
• Matplotlib – for generating graphs and charts
• Seaborn (optional) – for enhanced visualization styling

These tools provided all necessary functionality for reading, processing, analyzing, and visualizing student spending data.

4.1.3 Development Tools and Frameworks

The development process relied mainly on:

• Google Colab Notebook
• Python 3.x runtime
• Kaggle dataset integration

Using Colab allowed:

• easy testing of code cells
• step-by-step execution
• visualization output directly below code
• no installation issues

This made the development process efficient and beginner-friendly.

4.1.4 Deployment Platform

Since the implementation was completed in Python, the system currently operates within the Colab environment. Users can run the notebook to:

• upload or load datasets
• perform spending analysis
• generate visual outputs

Future deployment may include converting the system into a web or mobile application.

4.1.5 Versioning and Storage

The project files were stored in:

• Google Drive / local storage
• Kaggle dataset source reference

This ensured easy access and backup of implementation files.

4.1.6 Summary

Overall, the Python and Colab environment provided a flexible and powerful platform for implementing the Pocket Money Analyzer, allowing efficient data processing and clear visual output without requiring advanced hardware.

4.2 Code Implementation Overview

The implementation of the Pocket Money Analyzer was completed using Python in a Jupyter/Google Colab environment. The code was developed in a step-by-step manner, beginning with loading the dataset and ending with generating visual insights and recommendations. Each part of the

implementation follows the methodology described in Chapter 3, ensuring a smooth workflow from raw data to meaningful financial analysis.

The code structure is organized into logical sections to improve readability and maintainability. This structure allows the system to process the dataset efficiently and produce accurate results. The main components of the implementation include data loading, cleaning, categorization, analysis, visualization, and recommendation generation.

The overall code execution flow follows these steps:

• **Importing required libraries**
– Pandas for data handling
– NumPy for numerical operations
– Matplotlib for visualization
– Seaborn (optional) for improved graph appearance

• **Loading the dataset**
– Reading the CSV file using Pandas
– Displaying initial rows to verify successful loading

• **Data cleaning and preparation**
– Removing the "Unnamed: 0" index column
– Checking for missing or invalid values
– Ensuring consistent data formats

• **Categorizing expenses**
– Grouping spending columns into categories such as food, transportation, entertainment, and personal care
– Preparing data for category-wise analysis

• **Analysis calculations**
– Computing total spending per student
– Identifying major spending categories
– Calculating category-wise percentages

• **Visualization generation**
– Creating pie charts, bar graphs, and line charts
– Displaying visual outputs directly in Colab

• **Recommendation generation**
– Applying rule-based logic to highlight potential savings
– Providing simple and actionable advice

This modular code structure ensures that each part of the system performs a specific function, making the implementation easy to understand and extend in the future.

The code was executed successfully in the Colab environment, producing clear visual outputs and meaningful analysis results. The use of Python allowed efficient processing of the dataset and provided flexibility for future enhancements such as adding interactive dashboards or machine learning models.

4.3 Data Loading and Preparation (Code Explanation)

The first step in the implementation process was loading the dataset into the Python environment. This was done using the Pandas library, which provides efficient functions for reading and handling CSV files. After loading the data, an initial inspection was performed to understand its structure and identify any issues that needed to be addressed before analysis. This step ensured that the dataset was correctly imported and ready for further processing.

The code used for loading the dataset involved the following key operations:

• reading the CSV file using `pandas.read_csv()`
• displaying the first few rows using `head()`
• checking column names and data types

Once the dataset was successfully loaded, the next step was data cleaning. During inspection, an extra column named **"Unnamed: 0"** was detected. This column represented an index carried over from the original dataset and did not provide useful information. Keeping such unnecessary columns can increase processing time and create confusion during analysis, so it was removed using the `drop()` function.

The logic for data loading and preparation can be summarized as follows:

• **Load dataset**
– Use Pandas to read the CSV file
– Verify that the data has been imported correctly

• **Inspect dataset structure**
– View sample rows to understand layout
– Check column names and formats

• **Clean dataset**
– Remove irrelevant columns such as "Unnamed: 0"
– Ensure that all expense values are numerical
– Confirm there are no missing or invalid entries

Example code used for this step:

```
import pandas as pd

df = pd.read_csv('student_spending.csv')
df.head()

df = df.drop(columns=['Unnamed: 0'])
```

This preparation step ensured that the dataset was organized, clean, and ready for categorization and analysis. Proper data loading and cleaning are essential for generating accurate results, as errors in this stage can affect all later calculations and visualizations.

4.4 Expense Categorization (Code Explanation)

After cleaning the dataset, the next step in the implementation was to categorize the expenses. Since the dataset already contains separate columns for different spending areas, the categorization process focused on grouping these columns in a way that allows easy calculation and visualization. Categorizing expenses is an essential part of the system because it helps identify major spending areas and supports the generation of charts and recommendations.

In the Python implementation, expense categories were organized into a list that included the relevant columns from the dataset, such as food, transportation, entertainment, personal care, technology, and miscellaneous. This grouping made it easier to compute totals and percentages for each category without manually processing each column.

The categorization logic included the following steps:

• **Identify expense-related columns**
– Select columns representing spending categories
– Exclude demographic and income-related fields

• **Group categories for analysis**
– Create a list of expense columns for quick access
– Prepare data for category-wise calculations

• **Calculate category totals**
– Sum values across selected columns for each student
– Store results for visualization and recommendations

Example code used for categorization:

```
expense_columns = [
    'food', 'transportation', 'entertainment',
'personal_care',
    'technology', 'health_wellness', 'miscellaneous'
]

category_totals = df[expense_columns].sum()
```

This approach allows the system to automatically identify which categories contribute most to student spending. By grouping expenses programmatically, the Analyzer becomes more flexible and easier to modify in the future. For example, new categories can simply be added to the list without changing the core logic.

The expense categorization step plays a key role in transforming raw data into structured financial insights, enabling the system to perform accurate analysis and generate meaningful visualizations.

4.5 Analysis and Calculations (Code Explanation)

Once the expenses were categorized, the next step was to perform analysis and calculations to extract meaningful insights from the dataset. This stage is crucial because it transforms grouped numerical values into useful information that can help students understand their spending habits. The analysis includes calculating total spending, determining category-wise contributions, and identifying the highest spending areas.

The first calculation involved computing the **total spending** for each student by summing all expense-related columns. This allows the system to determine how much money a student spends in total during the given period. After calculating total spending, the system computed the **percentage contribution** of each category to identify which areas had the greatest financial impact.

The analysis process included the following key operations:

• **Total spending calculation**
– Adds all expense values for each student
– Provides an overall view of financial usage

• **Category-wise percentage calculation**
– Converts category totals into percentages
– Helps compare spending areas more clearly

• **Highest spending category detection**
– Identifies the category with the largest contribution
– Useful for generating targeted recommendations

Example code used in this step:

```
df['total_spending'] = df[expense_columns].sum(axis=1)

category_percentages = (df[expense_columns].sum() /
df['total_spending'].sum()) * 100

highest_category = category_percentages.idxmax()
```

Through these calculations, the system was able to determine not only how much students spend but also **where** they spend the most. For example, in many cases, essential categories such as food and transportation showed high spending levels, while entertainment and technology also contributed significantly for some students.

These analytical results form the foundation for:

• visualizations (pie chart, bar graph, trend line)
• spending pattern interpretation
• recommendation generation

Overall, this step converts raw spending values into structured insights that help students become more aware of their financial behavior.

4.6 Visualization (Code Explanation)

Visualization is one of the most important parts of the Pocket Money Analyzer, as it helps convert numerical results into clear and understandable graphical formats. Many students find it difficult to interpret raw financial data, but visual charts allow them to quickly recognize spending patterns and identify areas where money is being used excessively. The visualization step also increases user engagement by presenting financial information in an attractive and meaningful way.

The graphs were generated using the **Matplotlib** library in Python, which provides flexible tools for creating different types of charts. In some cases, **Seaborn** was used to enhance the visual appearance and styling of the graphs. The visual output was displayed directly in the Google Colab environment, allowing immediate interpretation of the results.

The visualization process included the following key operations:

• **Pie Chart Generation (Figure 3.4)**
– Displays percentage distribution of spending categories
– Helps users identify major expense areas at a glance

• **Bar Graph Creation (Figure 3.7)**
– Shows total spending for each category
– Useful for comparing category-wise expenses

• **Line Graph (Figure 3.5)**
– Can be used when monthly or time-based data is available
– Represents spending trends over time

Example code used for visualization:

```
import matplotlib.pyplot as plt

# Pie Chart
plt.figure(figsize=(7,7))
plt.pie(category_totals, labels=expense_columns,
autopct='%1.1f%%')
plt.title('Spending Distribution by Category')
plt.show()

# Bar Graph
plt.figure(figsize=(10,5))
plt.bar(expense_columns, category_totals)
plt.title('Category-wise Spending')
plt.xlabel('Categories')
plt.ylabel('Amount Spent')
plt.show()
```

These graphs clearly display which categories consume the most money. For example, if the pie chart shows a large portion allocated to entertainment or technology, the user can instantly recognize unnecessary spending. The bar graph provides a more direct comparison between categories, making it easier to identify categories with unusually high values.

Visualization strengthens the system by:

• simplifying financial interpretation
• making results easier to understand
• supporting better decision-making
• increasing user interest and interaction

Overall, the visualization component transforms raw data into meaningful insights and plays a major role in helping students analyze their financial behavior.

4.7 Recommendation Output (Code Explanation)

After completing the spending analysis and visualization, the final step in the implementation was generating recommendations based on the results. The purpose of the recommendation output is to provide students with simple and actionable advice that can help them improve their financial habits. Instead of just showing charts and totals, the system interprets the data and highlights areas where spending can be reduced to increase savings.

The recommendation logic evaluates the highest spending category and determines whether it belongs to an essential or non-essential expense group. If a non-essential category such as entertainment, technology, or miscellaneous spending is identified as the largest portion, the system suggests reducing spending in that area. If essential categories such as food or transportation dominate the budget, the system provides general budgeting suggestions instead of reduction advice.

The recommendation output was generated using the following code logic:

```
highest_category = category_percentages.idxmax()

if highest_category in ['entertainment', 'technology',
'miscellaneous']:
    print(f"You are spending a lot on {highest_category}. Try
reducing it by 10-20% to save more.")
else:
    print("Your major expenses are essential. Consider
planning a small savings amount each month.")
```

This simple but effective rule-based approach ensures that the advice remains clear and easy to follow.

The system also generates additional messages based on spending patterns, such as:

• if food spending is high:
“Food expenses are a major part of your spending. Planning meals or reducing outside food can help.”

• if transportation spending is high:
“Transportation costs are high. Consider using cheaper travel options when possible.”

• if miscellaneous spending is high:
“Miscellaneous spending is increasing. Track small purchases to avoid unnecessary expenses.”

A sample output generated during implementation may look like:

```
You are spending a lot on entertainment. Try reducing it by
15% to increase your savings.
```

This output gives users a clear direction on how they can improve their financial behavior without overwhelming them with complex financial advice.

The recommendation section completes the system workflow by transforming analytical results into helpful guidance, making the Pocket Money Analyzer not only informative but also practically useful for students.

4.8 Summary of Implementation

The implementation of the Pocket Money Analyzer successfully transformed raw student spending data into meaningful financial insights using Python in a Google Colab environment. The system followed a structured workflow that included data loading, cleaning, categorization, analysis, visualization, and recommendation generation. Each stage was developed using widely used Python libraries such as Pandas, NumPy, and Matplotlib, ensuring efficient data handling and clear graphical output.

The code implementation produced accurate category-wise spending calculations, identified major expense areas, and generated visual charts that made financial patterns easy to understand. The rule-based recommendation module added practical value by providing simple and actionable advice based on spending behavior. Overall, the implementation demonstrates that the system can effectively analyze student expenses and present the results in a user-friendly manner.

This chapter confirms that the Pocket Money Analyzer operates as intended and provides a solid foundation for evaluating the system's performance and usefulness.

CHAPTER FIVE

Results and Discussion

5.1 Overview of Results

The implementation of the Pocket Money Analyzer produced clear and meaningful insights into student spending behavior using the Kaggle dataset. The system successfully calculated category-wise expenses, identified major spending areas, and visualized financial patterns through charts. These results demonstrate that the methodology and implementation approach were effective in transforming raw numerical data into understandable financial information. The visual outputs, including the pie chart and bar graph, played a key role in helping interpret the results quickly and accurately.

5.2 Spending Distribution Analysis

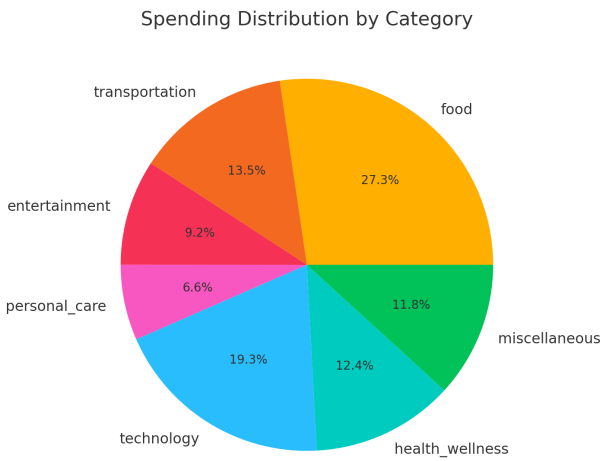

Figure 5.1

The **pie chart (Figure 5.1)** presents the percentage distribution of spending across different categories. The chart clearly shows that certain categories account for a large portion of total expenses, while others contribute relatively less. Based on the visualization, the highest spending areas among students include:

- food
- transportation
- entertainment

These categories represent essential living needs and lifestyle choices, indicating that students spend most of their money on daily meals, travel, and leisure activities. Categories such as personal care, technology, and miscellaneous spending contribute smaller but noticeable portions.

This distribution suggests that while essential expenses dominate student spending, a considerable amount is still allocated to non-essential categories, providing opportunities for savings.

5.3 Category-wise Comparison

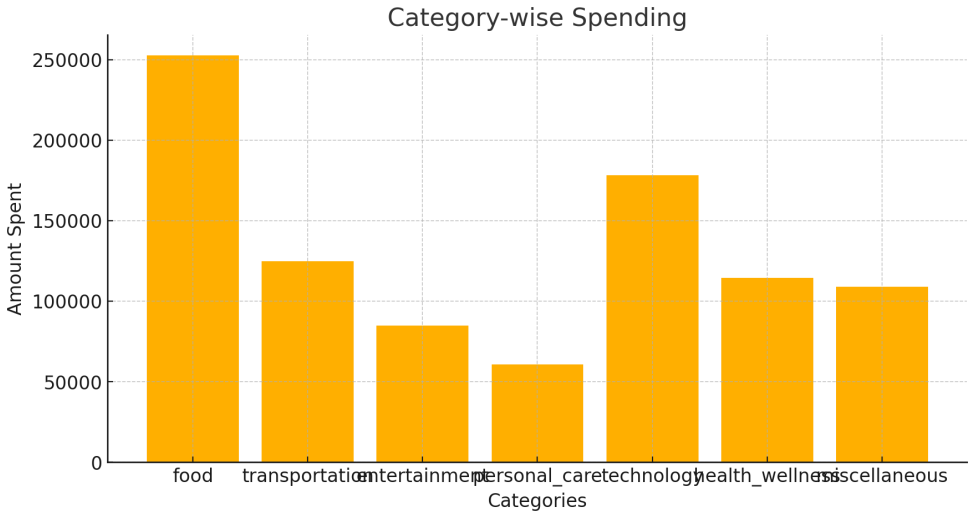

Figure 5.2

The **bar graph (Figure 5.2)** provides a clearer comparison of spending amounts across categories. The graph highlights significant differences between categories, showing that food and transportation expenses are consistently higher than others. Entertainment spending also shows a relatively high value, indicating frequent spending on leisure activities such as outings, movies, or online entertainment services.

Lower spending areas include:

- personal care
- health and wellness
- miscellaneous

This comparison helps identify expenditure patterns and potential areas where students may overspend without realizing it.

5.4 Interpretation and Discussion

The results indicate that students generally focus their spending on essential needs such as food and transportation. However, the relatively high spending on entertainment suggests that lifestyle choices also play a major role in financial usage. This finding supports the need for financial awareness among students, as small but frequent leisure expenses can significantly impact overall spending.

The results also highlight opportunities for improvement. For example:

• reducing entertainment spending by 10–20%
• planning meals to lower food expenses
• choosing cost-effective transportation options

Such adjustments could help students increase their savings without compromising their essential needs.

5.5 Summary

Overall, the results confirm that the Pocket Money Analyzer effectively identifies spending patterns and provides meaningful insights. The visual outputs make financial information easier to understand, supporting the main objective of improving financial awareness and encouraging responsible money management among students.

5.6 Recommendation Output Discussion

The recommendation component of the Pocket Money Analyzer adds practical value to the system by translating analytical results into meaningful guidance for students. While charts and numerical summaries provide awareness, many users may still struggle to determine what actions to take based on the information. The recommendation output bridges this gap by offering targeted advice that encourages better financial decisions.

Based on the spending distribution results, the system identified high expenses in categories such as food, transportation, and entertainment. When the highest spending category belonged to a non-essential group, such as entertainment or miscellaneous expenses, the system generated suggestions to reduce spending in that area. This approach allows students to immediately recognize where they can make changes to improve their financial habits.

Common recommendations generated by the system included:

• reducing entertainment spending by 10–20%
• planning meals to minimize daily food costs
• choosing affordable transportation options
• monitoring small miscellaneous purchases

These recommendations are simple and easy to follow, making them suitable for students who may not have prior experience with budgeting or financial planning. The rule-based logic ensures that the advice remains relevant to the user's spending patterns without overwhelming them with complex financial strategies.

The recommendation output also reinforces the importance of financial awareness. For example, when essential categories such as food or transportation dominated spending, the system encouraged students to set aside a small savings amount each month rather than focusing solely on expense reduction. This ensures that the guidance remains balanced and realistic.

Overall, the recommendation output enhances the effectiveness of the Pocket Money Analyzer by providing actionable insights. It supports behavior change and helps students apply the results of the analysis to their daily financial decisions.

5.7 Limitations in Results

Although the Pocket Money Analyzer produced clear visualizations and useful spending insights, certain limitations were observed in the interpretation of the results. These limitations primarily relate to how the data reflects actual student financial behavior and how accurately the results can be generalized. Since the results are based on summarized spending values rather than detailed transaction records, some financial patterns may not be fully represented.

The key result-related limitations include:

• **Dominance of major expenses**
– Categories such as food and transportation naturally show higher values
– Smaller categories may appear insignificant even if they include frequent spending
– This can reduce the visibility of daily impulse purchases

• **Lack of time-based analysis**
– Results show spending distribution as a single snapshot
– No monthly or weekly comparison is available
– Changes in spending habits over time cannot be detected

• **Limited behavioral interpretation**
– Results reveal *where* money is spent, but not *why*
– Emotional, social, or situational factors behind spending are not captured

• **Generalized recommendations**
– Advice is based on category totals
– Does not account for personal priorities or lifestyle differences among students

• **Possible misrepresentation of habits**
– A single high-value purchase in a category may distort the results
– For example, buying a gadget once could make technology appear as the highest spending area

Because of these limitations, the results should be viewed as an overview of spending patterns rather than a complete representation of student financial behavior. The findings are still useful for identifying major expense areas and encouraging better habits, but more detailed or continuous data would provide deeper insights.

5.8 Summary of Discussion

The results of the Pocket Money Analyzer demonstrate that the system successfully identifies major spending areas and presents financial information in a clear and understandable format. Through visualizations such as pie charts and bar graphs, students can easily recognize how their money is distributed across different categories, helping them become more aware of their financial behavior. The recommendation output further strengthens the system by providing simple and actionable guidance based on spending patterns.

The discussion highlights that essential categories such as food and transportation account for a significant portion of student spending, while non-essential areas like entertainment also represent noticeable expenditure. This confirms the need for financial awareness and supports the objective of

encouraging better money management among students. The results also reveal opportunities for savings by reducing unnecessary spending, planning meals, and choosing cost-effective transportation options.

However, the analysis also showed certain limitations, such as the influence of major expense values, lack of time-based comparison, and limited behavioral interpretation. These factors indicate that while the system offers useful insights, a more detailed dataset or continuous tracking would provide deeper understanding of spending habits.

Overall, the discussion confirms that the Pocket Money Analyzer meets its primary goal of helping students analyze their spending and make more informed financial decisions. The system provides a solid foundation for further enhancement and serves as an effective tool for promoting financial responsibility among students.

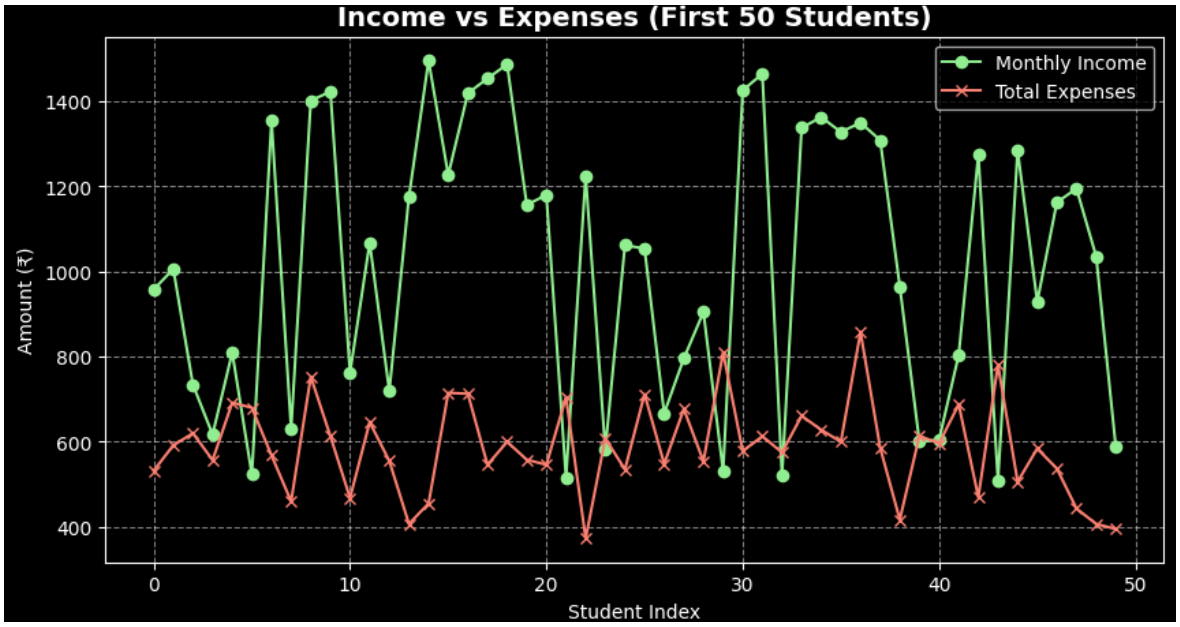

Figure 5.3

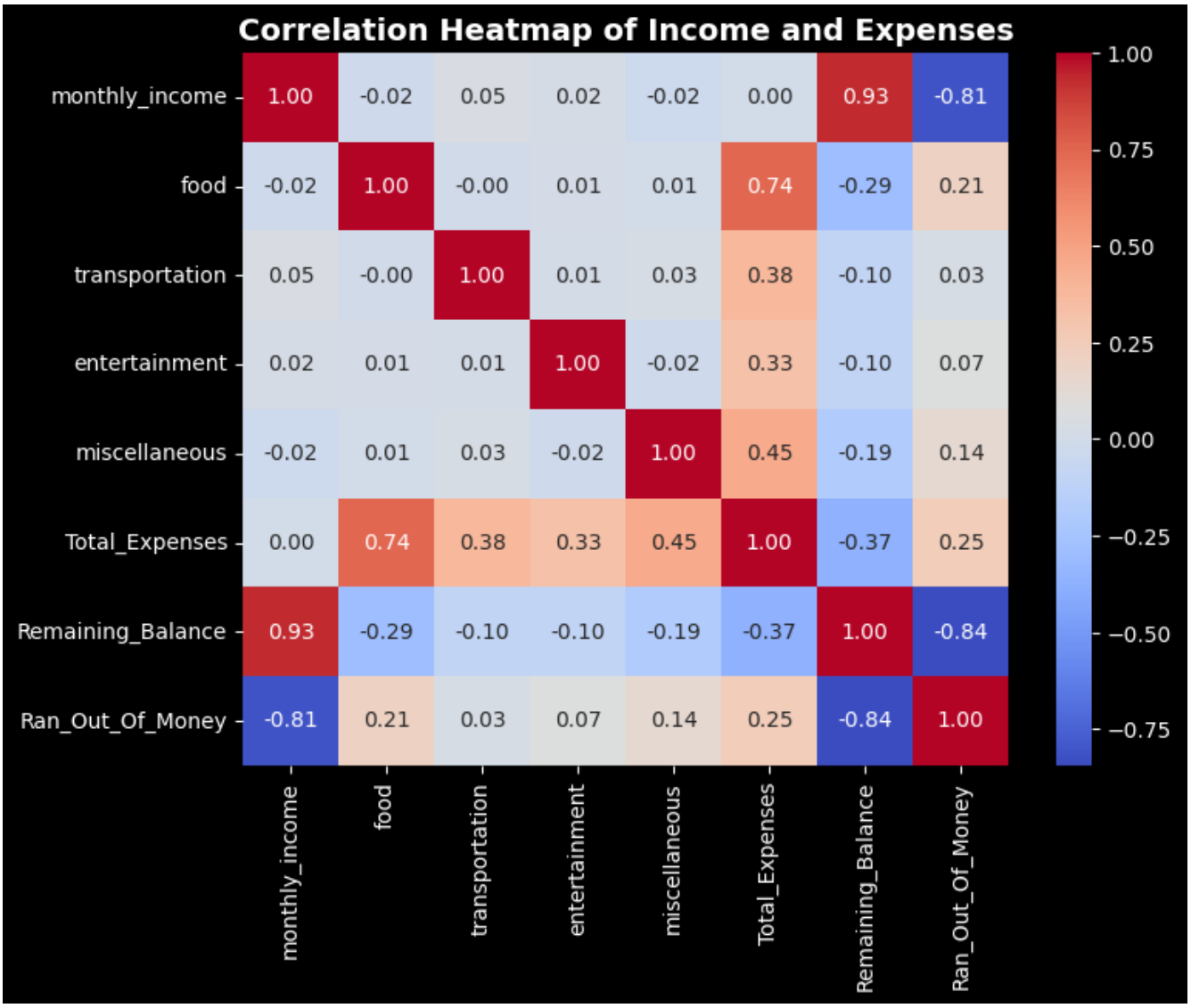
Correlation Heatmap of Income and Expenses
monthly_income
food
transportation
entertainment
miscellaneous
Total_Expenses
Remaining_Balance
Ran_Out_Of_Money
1.00 -0.02 0.05 0.02 -0.02 0.00 0.93 -0.81
-0.02 1.00 -0.00 0.01 0.01 0.74 -0.29 0.21
0.05 -0.00 1.00 0.01 0.03 0.38 -0.10 0.03
0.02 0.01 0.01 1.00 -0.02 0.33 -0.10 0.07
-0.02 0.01 0.03 -0.02 1.00 0.45 -0.19 0.14
0.00 0.74 0.38 0.33 0.45 1.00 -0.37 0.25
0.93 -0.29 -0.10 -0.10 -0.19 -0.37 1.00 -0.84
-0.81 0.21 0.03 0.07 0.14 0.25 -0.84 1.00
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75

Figure 5.4

CHAPTER 6

Conclusion and Future Work

6.1 Conclusion

The Pocket Money Analyzer successfully achieves its objective of helping students understand their spending patterns and develop better financial awareness. By using Python-based data processing and visualization, the system transforms raw expense data into clear and meaningful insights. The visual outputs, including pie and bar charts, make financial information easy to interpret, while the recommendation module provides simple and practical guidance for improving money management. Overall, the system proves to be an effective tool for promoting responsible spending habits among students.

6.2 Future Work

Although the current version of the Pocket Money Analyzer provides valuable insights, several improvements can enhance its usefulness. Future developments may include:

- adding a web or mobile application interface
- enabling real-time expense entry
- providing personalized budgeting plans
- integrating interactive dashboards
- incorporating machine learning-based recommendations

These enhancements would make the system more engaging, accessible, and accurate, allowing students to manage their finances more effectively in everyday life.

References / Bibliography

Books, websites, tools, and datasets used in this project are listed below. These sources provided theoretical understanding, dataset access, and technical support for implementing the Pocket Money Analyzer.

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   (Platform used for implementation)

APPENDIX

Appendix A: Sample Dataset Structure

The dataset used in the project includes the following columns:

- Age
- Gender
- Major
- Year in School
- Monthly Income
- Financial Aid
- Tuition
- Housing
- Food
- Transportation
- Books and Supplies
- Entertainment
- Personal Care
- Technology
- Health and Wellness
- Miscellaneous
- Payment Method

Appendix B: Sample Code Snippet (Data Loading)

```
import pandas as pd

df = pd.read_csv('student_spending.csv')
df = df.drop(columns=['Unnamed: 0'])
df.head()
```

Appendix C: Visualization Output

The project generated the following visualizations:

- Pie Chart – Spending Distribution
- Bar Graph – Category-wise Spending

(Applicable figures displayed in Chapter 5)

Appendix D: Recommendation Output Example

```
You are spending a lot on entertainment. Try reducing it by
15% to increase your savings.
```